

LAANE-NIGULA, Estonia

WMO: 261240

Lat: 58.9511N

Lon: 23.8156E

Elev: 82

StdP: 14.65

Time Zone: 2.00 (EUE)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-3.6	3.4	-8.0	3.6	-3.0	-1.6	5.1	4.4	22.8	34.0	21.0	33.3	4.2	100	0.610

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.9	81.6	67.0	78.0	65.0	74.7	63.5	69.2	77.7	67.1	75.1	65.2	72.3	7.6	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.2	96.8	73.3	64.2	90.3	70.4	62.4	84.4	68.8	33.3	77.8	31.7	75.2	30.2	72.5	78.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.8	18.3	16.3	-8.9	85.7	8.3	5.1	-14.8	89.3	-19.7	92.3	-24.4	95.2	-30.4	98.9		
(5)			-9.7	71.9	7.4	2.5	-15.1	73.7	-19.4	75.2	-23.6	76.6	-29.0	78.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.4	24.9	24.8	30.3	41.1	51.3	57.4	63.7	61.5	53.1	43.5	36.5	31.4
	DBStd	15.35	10.68	10.24	8.03	6.57	7.49	5.66	5.48	5.21	5.48	7.19	7.11	8.91
	HDD50	3683	780	705	610	276	70	5	0	1	33	221	405	577
	HDD65	7974	1245	1125	1075	716	429	237	91	135	357	667	855	1042
	CDD50	1274	0	0	0	10	111	228	424	357	126	18	0	0
	CDD65	93	0	0	0	0	6	11	50	26	0	0	0	0
	CDH74	865	0	0	0	0	72	122	460	209	2	0	0	0
	CDH80	168	0	0	0	0	7	14	108	39	0	0	0	0
Wind	WSAvg	7.8	8.6	8.5	8.4	7.9	7.3	7.6	6.5	6.7	6.8	7.7	8.6	9.5
Precipitation	PrecAvg	26.1	2.1	1.6	1.4	1.3	1.5	2.3	2.8	3.1	2.3	2.8	2.6	2.3
	PrecMax	38.0	4.4	4.1	2.6	2.8	2.9	4.1	5.2	6.7	4.7	5.4	5.2	6.3
	PrecMin	14.8	0.4	0.2	0.2	0.2	0.2	0.5	0.3	0.3	0.4	0.6	0.1	0.6
	PrecStd	4.7	0.9	0.8	0.7	0.8	0.7	1.0	1.3	1.4	1.1	1.3	1.2	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.1	41.6	55.4	69.5	80.2	81.9	87.0	84.6	71.7	61.5	50.3	48.1
		MCWB	42.1	39.2	43.6	54.2	63.2	66.0	69.0	68.7	61.1	55.6	48.5	46.4
	2%	DB	40.4	39.3	47.3	63.6	75.5	77.6	83.1	79.3	67.9	57.5	48.5	45.4
		MCWB	39.4	37.6	41.0	50.1	60.6	64.2	68.1	66.2	59.7	54.2	47.0	43.8
	5%	DB	38.2	37.6	42.5	58.7	70.8	73.2	79.1	75.4	65.1	55.4	47.2	43.7
		MCWB	37.1	36.3	38.3	47.4	57.6	61.3	66.3	64.3	58.4	53.1	45.9	42.3
	10%	DB	36.6	36.1	39.5	54.3	65.9	69.8	75.5	72.1	62.5	53.4	45.7	41.6
		MCWB	35.6	35.0	36.5	45.3	55.7	59.2	65.0	63.0	57.4	51.4	44.5	40.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.2	40.0	45.7	55.8	66.9	69.5	72.1	71.1	63.2	56.7	49.1	46.6
		MCDB	42.8	40.8	52.5	66.8	74.5	78.1	81.7	79.6	68.0	58.7	49.9	47.9
	2%	WB	39.5	38.0	41.5	51.2	63.0	65.7	69.9	68.1	61.1	55.0	47.5	44.3
		MCDB	40.0	38.9	46.0	60.3	71.9	74.0	79.0	76.1	65.8	56.9	48.3	45.3
	5%	WB	37.2	36.6	38.8	48.7	59.9	63.0	68.1	66.0	59.6	53.5	46.0	42.6
		MCDB	37.9	37.5	41.8	57.4	68.2	71.5	76.4	73.1	63.6	55.1	46.9	43.5
	10%	WB	35.6	35.1	37.0	46.1	56.2	60.6	66.2	64.2	58.0	51.8	44.7	40.5
		MCDB	36.4	36.0	39.2	52.9	64.1	68.1	74.3	70.1	61.6	53.3	45.6	41.5
Mean Daily Temperature Range	5% DB	MDBR	8.5	9.4	12.3	16.8	19.2	17.2	16.9	17.3	15.0	11.1	6.8	6.9
		MCDBR	6.7	7.3	16.1	26.7	26.1	24.2	23.3	22.2	18.8	12.6	7.2	6.2
		MCWBR	6.8	6.8	10.9	15.1	13.1	12.3	10.3	11.2	11.4	9.1	6.7	6.0
	5% WB	MCDBR	6.8	7.1	14.3	23.8	22.0	20.5	20.3	18.9	15.7	10.4	6.7	5.5
		MCWBR	6.9	6.6	10.2	14.3	12.0	11.5	10.0	10.6	10.4	8.3	6.6	5.7
Clear-Sky Solar Irradiance	taub	0.282	0.310	0.334	0.372	0.366	0.364	0.392	0.386	0.354	0.336	0.317	0.274	
	taud	2.156	2.195	2.260	2.318	2.404	2.439	2.382	2.395	2.469	2.407	2.228	2.119	
	Ebn at Noon	164	220	253	264	274	277	264	257	248	213	149	123	
	Edh at Noon	15	24	30	34	33	33	34	31	25	20	14	11	
All-Sky Solar Radiation	RadAvg	89	266	748	1266	1775	1866	1756	1384	856	381	108	58	
	RadStd	15	48	84	139	171	113	144	130	102	49	20	8	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (51 neighbors)		+0.70	N/A	N/A	N/A	N/A	N/A	-239	N/A	N/A	N/A

Nomenclature: See separate page